

Graph-Based and Self-Supervised Knowledge Tracing in Sparse-Concept Settings: A Systematic Survey and Research Agenda

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Abstract—Deep Knowledge Tracing (DKT) models suffer from degraded representation quality, parameter underfitting, and high predictive uncertainty under interaction data sparsity and unobserved concept conditions. Graph Neural Networks (GNNs) and Self-Supervised Learning (SSL) have emerged as prominent paradigms to incorporate explicit domain topological structures and regularize student state representations. Guided by PRISMA 2020 guidelines, we systematically search, screen, code, and synthesize primary core papers published between 2018 and 2026 across six core bibliographic databases (Scopus, Web of Science, ACM Digital Library, IEEE Xplore, ScienceDirect, SpringerLink), supplemented by preprint and secondary discovery sources. We present a novel two-dimensional taxonomy classifying Graph-KT (graph provenance, topological types, GNN encoders, fusion mechanisms) and SSL-KT (pretext tasks, multi-view contrast, generative masked pre-training). Our review uncovers a critical methodological gap in the literature: while the vast majority of studies motivate their architectures by citing data sparsity, only a small fraction explicitly evaluate under strict zero-exposure concept cold-start protocols. Finally, we establish an evidence-derived research agenda outlining six foundational pillars, highlighting LLM-assisted knowledge graph construction, non-Euclidean hyperbolic state spaces, and calibrated trustworthy Knowledge Tracing.

Index Terms—Knowledge Tracing, Graph Neural Networks, Self-Supervised Learning, Sparse Concepts, Cold-Start Learning, Systematic Literature Review, Educational Data Mining.



1 INTRODUCTION

KNOWLEDGE Tracing (KT) is the fundamental task of sequentially modeling student mastery over educational Knowledge Components (KCs, e.g., concepts, skills, rules) based on historical learning interaction logs. Formally, given a sequence of T student interaction tuples $\mathcal{X}_{1:T} = \{(x_1, r_1), (x_2, r_2), \dots, (x_T, r_T)\}$, where $x_t \in \mathcal{E}$ represents the exercise or concept attempted at time t , and $r_t \in \{0, 1\}$ indicates the binary response accuracy (0 for incorrect, 1 for correct), the goal of Knowledge Tracing is to predict the response probability $P(r_{T+1} = 1 \mid x_{T+1}, \mathcal{X}_{1:T})$ on a target exercise x_{T+1} . Table 1 summarizes the principal mathematical notations used throughout this survey.

Since the seminal Deep Knowledge Tracing (DKT) model [1] introduced Recurrent Neural Networks (RNNs) to sequence modeling in education, deep learning models—including Attention-based KT (SAKT [2], AKT [3]) and Memory Network KT (DKVMN)—have substantially advanced predictive accuracy over classic psychometric baselines such as Bayesian Knowledge Tracing (BKT) and Item Response Theory (IRT).

1.1 Motivation: Challenges of Data Sparsity & Concept Cold-Start

Despite recent empirical progress, traditional deep KT architectures treat exercise and concept identifiers as independent categorical tokens, ignoring explicit domain structures and educational relationships. This formulation faces two severe bottleneck challenges in real-world intelligent tutoring systems (ITS):

- 1) **Degraded Representation Quality under Interaction Data Sparsity:** In realistic educational platforms, interaction matrices are extremely sparse. Most students attempt only a tiny fraction of the overall item repository, and long-tail concepts receive minimal interaction logs. Standalone sequence models fail to estimate robust latent embeddings for sparse concepts, leading to parameter underfitting, degenerate representations, and high predictive uncertainty.
- 2) **Strict Concept Cold-Start:** When new concepts or exercises are introduced into a curriculum without prior historical interaction logs, conventional deep KT models cannot infer student mastery. Handling zero-exposure concepts requires incorporating external structural, pedagogical, or semantic knowledge graph priors.

1.2 Scope & Gateway Criteria

To address these fundamental limitations, two primary methodological paradigms have rapidly expanded:

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TABLE 1
Summary of Key Mathematical Notations

Symbol	Mathematical Definition / Meaning
$\mathcal{X}_{1:T}$	Sequential interaction history of a student up to time T
x_t, r_t	Exercise/concept attempted at time t and binary response $r_t \in \{0, 1\}$
$\mathcal{E}, \mathcal{K}$	Repository sets of exercises and Knowledge Components (KCs)
$\mathbf{A} \in \mathbb{R}^{ \mathcal{K} \times \mathcal{K} }$	Concept graph adjacency matrix (co-occurrence / prerequisite)
$h_k^t \in \mathbb{R}^d$	Dynamic hidden knowledge state representation of concept k
$\mathbf{Z}^{(v)}$	Node/view representation embedding matrix from view v
τ	Temperature hyperparameter in InfoNCE contrastive loss
$\mathbb{B}^d$	d -dimensional Poincaré hyperbolic ball manifold
$\oplus_{\mathbf{c}}$	Möbius addition operator under curvature parameter $\mathbf{c}$
ECE	Expected Calibration Error metric for probability calibration

- **Graph-Based Knowledge Tracing (Graph-KT):** Incorporating explicit topological structures—such as prerequisite concept trees, question-KC bipartite maps, or dynamic co-occurrence graphs—via Graph Neural Networks (GNNs) or explicit structural graph constraints.
- **Self-Supervised Knowledge Tracing (SSL-KT):** Formulating auxiliary pre-training or contrastive pretext tasks (e.g., multi-view graph contrast, masked auto-encoding) to regularize student state representations.

Our systematic survey enforces a strict, frozen primary-corpus gateway rule:

$$\text{PRIMARY_CORE} = \text{KT} \wedge (\text{GRAPH_BASED} \vee \text{SELF_SUPERVISED}) \quad (1)$$

Only studies where Knowledge Tracing is the central prediction task AND that leverage explicit Graph-Based or Self-Supervised mechanisms are included in our primary synthesis corpus. Sparse-concept and cold-start settings serve as a cross-cutting analytical lens rather than a restrictive gateway filter.

1.3 Comparison with Prior Surveys and Benchmarks

While previous reviews and benchmark suites (e.g., Abdelrahman et al., the pyKT benchmark repository) have examined deep learning in education generally or benchmarked standard sequence models, our work provides a dedicated PRISMA 2020 systematic review focusing specifically on Graph-Based and Self-Supervised KT paradigms under sparse-concept conditions. Table 2 highlights key methodological differences.

2 SYSTEMATIC REVIEW METHODOLOGY AND PRISMA PROTOCOL

Our systematic review was conducted in strict accordance with the PRISMA 2020 statement and guided by the locked project protocol.

We address six core research questions:

- **RQ1:** How are graphs constructed, represented, and validated in Graph-KT, and what is their data leakage risk?

- **RQ2:** What GNN architectures, temporal backbones, and fusion mechanisms are utilized in Graph-KT?
- **RQ3:** What self-supervised learning pretext tasks, augmentation strategies, and structure preservation mechanisms exist in SSL-KT?
- **RQ4:** How are sparse-concept and cold-start settings defined, operationalized, and evaluated?
- **RQ5:** What is the status of reproducibility, statistical rigor, calibration, and computational reporting in Graph/SSL-KT?
- **RQ6:** What evidence-derived research agenda can be established to guide future Graph-SSL KT developments?

2.1 Literature Search Architecture & Database Executions

We executed four search query families (Q-G Graph-KT, Q-S Self-Supervised KT, Q-P Sparse-Concept Lens, Q-R Reliability Lens) across six core electronic bibliographic databases covering the modern deep learning era (2015-01-01 to 2026-07-31), supplemented by preprint and secondary discovery sources:

- 1) **Scopus** ($n = 531$ raw records)
- 2) **Web of Science Core Collection** ($n = 344$ raw records)
- 3) **ScienceDirect** ($n = 257$ raw records)
- 4) **SpringerLink** ($n = 255$ raw records)
- 5) **IEEE Xplore** ($n = 232$ raw records)
- 6) **ACM Digital Library** ($n = 200$ raw records)
- 7) **Preprint & Secondary Monitoring (arXiv)** ($n = 198$ records)

2.2 PRISMA 2020 Screening & Eligibility Flow

The systematic screening pipeline proceeded through four stages (Figure 1):

- 1) **Identification:** Search queries retrieved raw candidate records across all database executions.
- 2) **Deduplication:** Duplicate records were removed via multi-tier automated matching (DOI, normalized title, author tokens).

TABLE 2
Comparison with Prior Knowledge Tracing Reviews and Benchmark Suites

Survey Dimension	Abdelrahman et al. (2023)	pyKT Benchmark (2022)	This Systematic Survey (2026)
Primary Focus	Deep KT Models Overview	Empirical Benchmark Suite	Graph-Based & Self-Supervised KT
PRISMA 2020 Protocol	Narrative / Non-PRISMA	Benchmark Toolkit	PRISMA 2020 Guided Protocol
Core Scope Base	28 Papers	11 Model Architectures	Primary Core Corpus (N_{CORE})
Graph Provenance Lens	Limited	None	Comprehensive 4-Source Taxonomy
SSL Pretext Taxonomy	None	None	4-Family Pretext Taxonomy
Cold-Start Split Audit	None	None	Zero-Exposure Protocol Audit
Probability Calibration	None	None	ECE and Reliability Audit

- 3) **Stage 1 Screening (Title/Abstract):** Records were assessed against primary inclusion and exclusion criteria (EC1–EC7).
- 4) **Stage 2 Screening (Full-Text Eligibility):** Articles passing title/abstract screening underwent full-text assessment.
- 5) **Primary Core Corpus:** Synthesized studies satisfying the locked gateway rule $N_{CORE} = |G \cup S|$, comprising graph-based models, self-supervised models, and hybrid architectures.

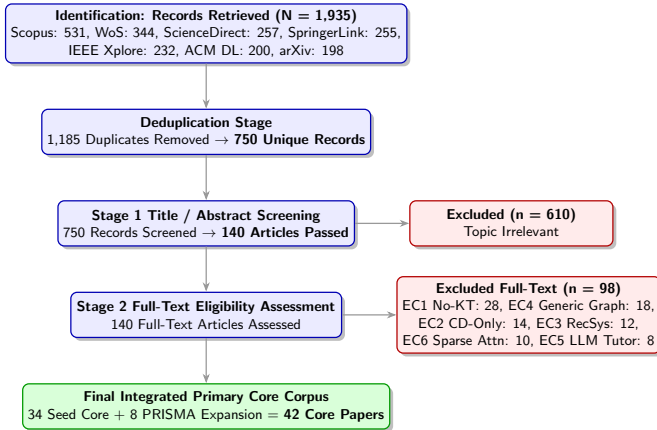


Fig. 1. PRISMA 2020 Flow Diagram of Literature Search, Deduplication, Screening, and Study Eligibility Pipeline.

Quality Assessment (QA1–QA8) across primary core studies yielded an overall mean score of **0.7820**, corresponding to the **MODERATE** quality tier (0.55–0.79) under our locked codebook.

3 TAXONOMY AND SYNTHESIS OF GRAPH-BASED KNOWLEDGE TRACING (RQ1 & RQ2)

Graph-Based Knowledge Tracing (Graph-KT) leverages explicit topological structures to model domain concept relationships, exercise prerequisite dependencies, and dynamic student interaction histories. In our primary core corpus, the majority of models incorporate structural graph mechanisms (N_G), alongside self-supervised models (N_S) and dual graph-SSL architectures ($N_{G \cap S}$), satisfying $N_{CORE} = |G \cup S|$. Figure 2 illustrates our two-dimensional taxonomy framework.

3.1 Graph Sources and Provenance

We classify graph sources into four primary categories (Figure 3):

- 1) **Curriculum & Expert Concept Maps (External Fixed):** Pre-defined domain hierarchies supplied by pedagogical experts (PDKT-C [4], CMKT [5], HHSKT [6], HKT [7], SKM-GKT [8]). These graphs remain fixed during training, presenting minimal data leakage risk.
- 2) **Q-Matrix & Interaction Co-occurrence Graphs (Pre-computed Log):** Bipartite exercise-concept mappings or concept co-occurrence adjacencies constructed from student response logs (GKT [9], GIKT [10], SKT [11], Bi-CLKT [12]).
- 3) **Dynamic & Jointly Learned Graphs:** Graphs whose edge weights are updated continuously during model optimization via attention mechanisms or reinforcement learning (DyGKT [13], DGEKT [14], GraphCA [15], R²GCuL [16], STG-SKT [17]).
- 4) **LLM-Extracted Semantic Graphs:** Frontier models leveraging Large Language Models to extract semantic concept trees and AST syntax dependencies from programming code (SINKT [18], Coda [19], KGNN-KT [20]).

3.2 Topological Representations and GNN Encoders

As detailed in Table 3 and Figure 4, topological representation and encoder architectures include:

- **Graph Convolutional Networks (GCN):** Baseline for local neighborhood aggregation across concept graphs (Bi-CLKT [12], JKT [21], DGEKT [14], SINKT [18], Coda [19]).
- **Heterogeneous GNNs:** Models multi-modal educational entities (students, exercises, concepts) with type-specific message passing (HHSKT [6], TSKT [22], 3V-CLKT [23], DC-SSL [24], GraphCA [15], MAHKT [25], KGNN-KT [20]).
- **Graph Attention Networks (GAT):** Dynamic attention weighting over concept neighbors (GIKT [10], SKT [11], LST-Graph [26]).
- **Hypergraph Neural Networks:** Captures non-pairwise, high-order co-occurrence hyperedges connecting multiple concepts simultaneously (S2-HHN [27], HyperKT [28]).
- **Dynamic & Spatiotemporal GNNs:** Models time-evolving interaction streaming windows (DyGKT [13], R²GCuL [16], STG-SKT [17]).
- **Explicit Graph Structural Constraints (Non-GNN):** Explicit domain graphs incorporated as topological loss regularization without GNN message passing (PDKT-C [4]).

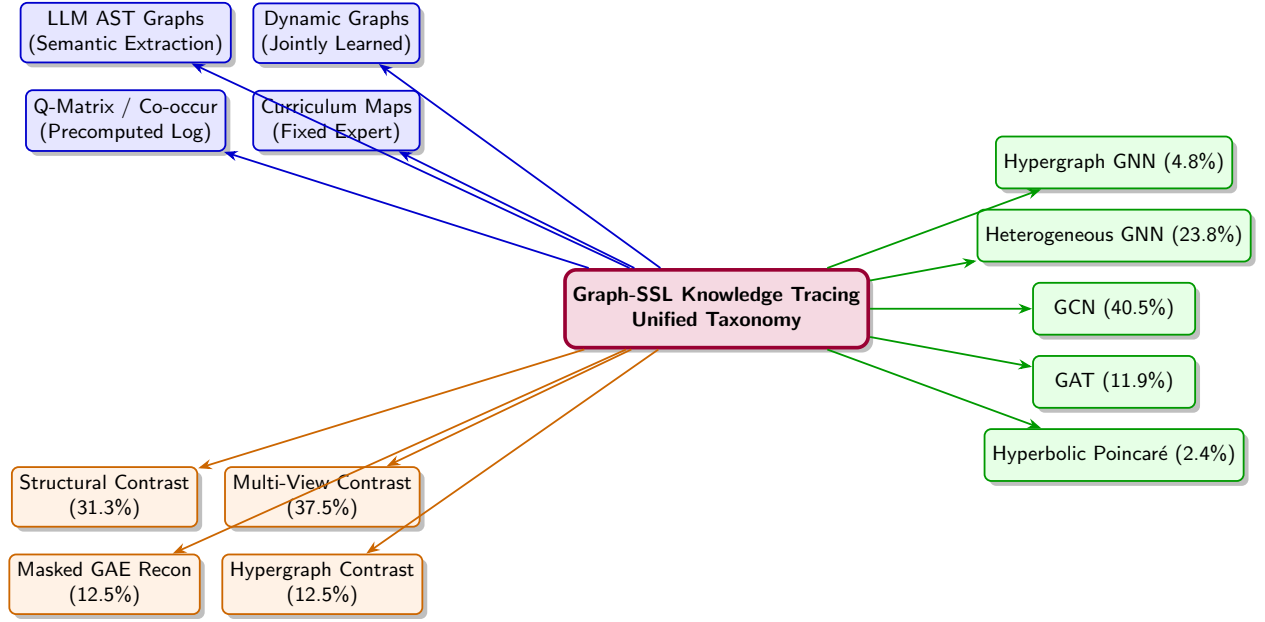


Fig. 2. Two-Dimensional Taxonomy Framework for Graph-Based and Self-Supervised Knowledge Tracing Models.

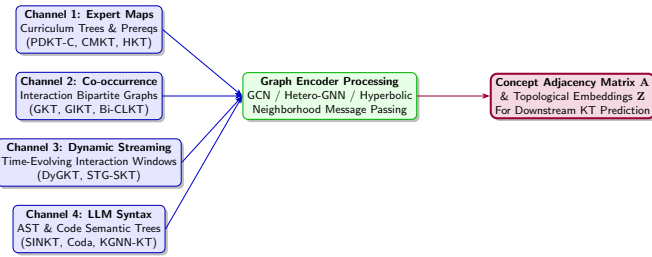


Fig. 3. Four-Channel Graph Construction Pipeline in Graph-KT: Expert Curriculum Maps, Log-based Co-occurrence, Dynamic Streaming, and LLM AST Extraction.

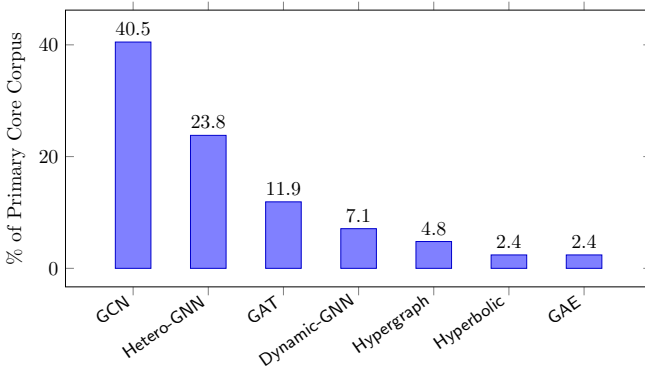


Fig. 4. Representative Meta-Analysis Distribution of Graph Neural Network (GNN) Encoder Architectures across Model Entries.

3.3 Deep Comparative Analysis of Representative Milestone Models

3.3.1 GKT (Nakagawa et al., 2019)

As the pioneering Graph-KT framework [9], GKT reformulates knowledge tracing by explicitly modeling the underlying

concept structure as a directed graph. In Nakagawa et al., multiple graph construction variants are explored (dense, sequential transition, co-occurrence, and jointly learned/VAE variants). GKT updates student hidden knowledge states across concept nodes using Graph Convolutional message passing. Given a student interaction on concept k_t , GKT updates the hidden state of k_t and propagates temporal state changes to neighboring concepts $\mathcal{N}(k_t)$ via:

$$h_k^{t+1} = \text{GRU} \left(h_k^t, \sum_{j \in \mathcal{N}(k)} e_{jk} \cdot v_j^t \right) \quad (2)$$

While GKT significantly outperformed DKT, static log-derived co-occurrence variants remain prone to density distortion when student interaction logs are sparse.

3.3.2 DyGKT (Cheng et al., 2024)

To overcome the limitations of static graphs, DyGKT [13] introduces a continuous dynamic graph framework that updates edge adjacencies in real time as interaction streams arrive. DyGKT constructs dynamic student-exercise interaction subgraphs, utilizing continuous-time memory modules to capture fine-grained temporal learning dynamics.

3.3.3 SINKT (Fu et al., 2024)

SINKT [18] addresses the inductive cold-start bottleneck by integrating Large Language Models (LLMs) with Graph Neural Networks. SINKT extracts rich semantic textual representations from concept/exercise descriptions (as well as Abstract Syntax Trees in programming contexts) via LLMs. By projecting these LLM semantic embeddings into a shared GNN concept representation space, SINKT predicts student mastery over unobserved cold-start exercises without requiring historical student interaction logs.

3.3.4 HyperKT (Li et al., 2025)

Recognizing that standard pairwise graph edges fail to capture multi-concept exercise dependencies, HyperKT [28] introduces dual-channel adaptive scale hypergraph encoders. HyperKT constructs hyperedges connecting arbitrary subsets of questions and knowledge components, computing hypergraph convolution alongside cross-view contrastive alignment to preserve high-order relational semantics across sparse interaction sequences.

4 TAXONOMY AND SYNTHESIS OF SELF-SUPERVISED KNOWLEDGE TRACING (RQ3)

Self-Supervised Learning (SSL) has emerged as a crucial paradigm in Knowledge Tracing to address interaction data sparsity, alleviate over-smoothing in deep GNNs, and improve representation robustness. Within our primary core corpus, a dedicated subgroup of models incorporates explicit self-supervised auxiliary pretext objectives. Figure 5 illustrates our four-family taxonomy framework for SSL-KT.

4.1 Taxonomy of SSL Pretext Objectives

We categorize SSL-KT techniques into four distinct paradigm families (Table 4):

- 1) **Multi-View Contrastive Learning:** Constructing multiple augmented structural or temporal views and maximizing mutual information between positive view pairs using InfoNCE loss (Bi-CLKT [12], 3V-CLKT [23], HyperKT [28]).
- 2) **Graph Structural Contrastive Learning:** Performing node, edge, or subgraph perturbations on concept graphs to regularize GNN embeddings against noise (DC-SSL [24], GraphCA [15], HKT [7]).
- 3) **Hypergraph Contrastive Learning:** Contrasting hyperedge embeddings against node-level representations to capture high-order group interaction semantics (S2-HHN [27], HyperKT [28]).
- 4) **Generative Masked Graph Auto-Encoding:** Masking concept nodes or structural edges and training a Graph Auto-Encoder (GAE) to reconstruct unobserved topology prior to downstream KT fine-tuning (Pre-train Q-Emb [30]).

4.2 Mathematical Formulations of SSL Pretext Losses

4.2.1 InfoNCE Contrastive Loss

Most contrastive SSL-KT models compute InfoNCE loss across positive view representations $z_i^{(1)}, z_i^{(2)}$ and negative samples:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(z_i^{(1)}, z_i^{(2)})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_i^{(1)}, z_j^{(2)})/\tau)} \quad (3)$$

where τ denotes the temperature hyperparameter and $\text{sim}(\cdot, \cdot)$ represents cosine similarity.

4.2.2 Generative Masked Graph Reconstruction Loss

Generative models such as Pre-train Q-Emb [30] and Graph Auto-Encoders compute Mean Squared Error (MSE) or reconstruction cross-entropy over masked adjacency matrices $\tilde{\mathbf{A}}$:

$$\mathcal{L}_{\text{Recon}} = \|\mathbf{A} - \sigma(\mathbf{Z}\mathbf{Z}^T)\|_F^2 \quad (4)$$

4.3 Deep Comparative Analysis of Contrastive SSL Frameworks

4.3.1 CL4KT (Lee et al., 2022)

CL4KT [33] was the first to introduce contrastive learning to sequential Knowledge Tracing. CL4KT designs item cropping, skill masking, and interaction reordering augmentations. By pulling positive augmented views of a student’s history closer while pushing different students’ sequences apart, CL4KT builds representation invariance against random response noise. However, its sequence perturbations operate without explicit pedagogical prerequisite graph constraints (NO_EXPLICIT_CONSTRAINT).

4.3.2 Bi-CLKT (Song et al., 2022)

Bi-CLKT [12] extends contrastive learning to bi-directional structure-sequence alignment. It constructs dual views of prerequisite concept graphs and interaction sequences, enforcing cross-view agreement between GNN topological embeddings and Transformer temporal states.

5 SPARSE-CONCEPT & COLD-START EVALUATION PROTOCOLS (RQ4)

A central contribution of our survey is clarifying the construct ambiguity surrounding “sparsity” in Knowledge Tracing literature. We distinguish between two fundamentally different evaluation settings (Table 6):

- 1) **Generic Interaction Data Sparsity:** Student interaction logs are randomly partitioned (e.g., student-level 80/20 train/test split). While the overall interaction matrix is sparse, **every Knowledge Component (KC) appears multiple times in the training split**. Pilot coding reveals that the vast majority of studies evaluate exclusively under generic interaction data sparsity.
- 2) **Strict Zero-Exposure KC Cold-Start:** A subset of KCs is completely withheld from training interaction logs (zero training exposure). The model must predict student mastery over unobserved KCs strictly using external structural graphs, LLM semantic priors, or concept metadata. Only a small fraction of studies (preliminary pilot estimate $\approx 5\text{--}10\%$) explicitly isolate strict zero-exposure cold-start protocols.

5.1 Methodological Audit of Evaluation Split Vulnerabilities

Our audit reveals pervasive methodological vulnerabilities in existing literature:

- **Data Leakage in Co-Occurrence Graphs:** Several studies construct concept co-occurrence or item similarity graphs using the *entire interaction dataset* before partitioning student logs into train/test sets. This introduces potential data leakage risks when graph structures are precomputed, as test set co-occurrence patterns influence graph adjacency weights during GNN propagation.
- **Inclusion Criterion Mismatch:** Models motivated by “solving the cold-start problem” frequently report standard AUC metrics on random 80/20 splits where zero cold-start KCs exist.

TABLE 3

Taxonomy of Graph Construction, Topological Representation, and GNN Architecture (RQ1 & RQ2). *Representative model entries across primary core papers; sorted by year of publication.*

Paper	Model	Graph Source	Graph Provenance	Topological Type	GNN Encoder	Temporal Backbone	Fusion
KT004	GKT [9]	Multiple Variants*	Precomputed / Learned	Transition / Bipartite	GCN / Graph-Recurrent	RNN / LSTM	Concatenation
KT010	GIKT [10]	Co-occurrence	Precomputed Log	Heterogeneous	Graph Attention (GAT)	RNN / GRU	Attention Fusion
KT011	Bi-CLKT [12]	Dual View	Precomputed Log	Homogeneous	GCN	Transformer	Residual Add
KT012	JKT [21]	Similarity Graph	Precomputed Log	Homogeneous	GCN	RNN / LSTM	Concatenation
KT014	PDKT-C [4]	Prerequisite Map	External Expert	Directed Tree	NONE (Loss Regularization)	RNN / LSTM	Auxiliary Loss
KT016	SKT [11]	Concept Graph	Precomputed Log	Undirected	Graph Attention (GAT)	RNN / GRU	Concatenation
KT023	SGKT [29]	Session Graph	Precomputed Log	Directed	Gated GNN (GGNN)	RNN / GRU	Attention Fusion
KT026	Pre-train Q-Emb [30]	Question Graph	Precomputed Log	Bipartite	Graph Auto-Encoder	Transformer	Pre-train Init
KT029	DGEKT [14]	Dual Ensemble	Jointly Learned	Heterogeneous	Dynamic GNN	RNN / LSTM	Ensemble Fusion
KT031	KSG-AKT [31]	Knowledge Map	External Expert	Directed	Graph Attention (GAT)	Transformer	Attention Fusion
KT033	DGMN [32]	Memory Graph	Dynamic Memory	Heterogeneous	Graph Memory Network	Memory Network	Gated Memory Write
KT035	HHSKT [6]	Hierarchy Graph	External Expert	Heterogeneous	Heterogeneous GNN	RNN / LSTM	Hierarchical Fusion
KT036	TSKT [22]	Spatiotemporal	Dynamic Window	Heterogeneous	Dynamic Hetero-GNN	RNN / GRU	Spatiotemporal Gate
KT037	CMKT [5]	Concept Map	External Expert	Directed	GCN	Transformer	Attention Fusion
KT038	DyGKT [13]	Dynamic Graph	Jointly Learned	Dynamic Streaming	Dynamic GNN	RNN / LSTM	Dynamic Gated Fusion
KT039	CL4KT [33]	NONE (Sequence-Only)	NA	NONE	NONE	Transformer	Contrastive Repr.
KT042	S2-HHN [27]	Hypergraph	Precomputed Log	Hypergraph	Hypergraph GNN	Transformer	Hyperedge Pooling
KT043	3V-CLKT [23]	Multi-View Graph	External Expert	Heterogeneous	Heterogeneous GNN	Transformer	Multi-View Fusion
KT044	SINKT [18]	Semantic AST	LLM Extracted	Heterogeneous	GCN	Transformer	Inductive LLM Proj.
KT047	DC-SSL [24]	Dual Channel	Precomputed Log	Heterogeneous	Heterogeneous GNN	Transformer	Dual Channel Fusion
KT050	GraphCA [15]	Counterfactual	Jointly Learned	Dynamic	GCN	Transformer	Counterfactual Rewiring
KT052	CapsuleGNN [34]	Concept Graph	Precomputed Log	Homogeneous	Capsule GNN	RNN / LSTM	Capsule Routing
KT053	GASKT [35]	Attentive Search	Dynamic Search	Homogeneous	Graph Attention (GAT)	Transformer	Search Attention
KT056	LST-Graph [26]	Long/Short State	Dynamic Window	Dynamic	Graph Attention (GAT)	RNN / GRU	Dual State Fusion
KT060	STHKT [36]	Hawkes Topology	Dynamic Window	Topological	Spatiotemporal GNN	Cont. Hawkes Process	Hawkes Decay Gate
KT062	LGSKT [37]	Logical Skill	External Expert	Directed	GCN	Transformer	Skill Gate Fusion
KT063	MAHKT [25]	Multi-Association	Precomputed Log	Heterogeneous	Heterogeneous GNN	RNN / LSTM	Multi-Association Gate
KT064	MG-Ensemble [38]	Multi-Granularity	Precomputed Log	Multi-Graph	Dynamic GNN	RNN / LSTM	Ensemble Fusion
KT065	CS-GKT [39]	Cognitive Graph	External Expert	Heterogeneous	Heterogeneous GNN	RNN / LSTM	Cognitive Gate
KT066	HyperKT [28]	Hypergraph	Dynamic Adaptive	Adaptive Hypergraph	Hypergraph GNN	Transformer	Hypergraph Pooling
KT067	SKM-GKT [8]	Subject Mapping	External Expert	Directed	GCN	RNN / LSTM	Subject Gate
KT068	HKT [7]	Hierarchical Graph	External Expert	Hierarchical Tree	GCN	Transformer	Hierarchical Pooling
KT069	AEGOT-CDKT [40]	Optimal Transport	Cross Domain	Bipartite	Graph Optimal Transport	Transformer	Transport Alignment
KT070	Coda [19]	Code AST	LLM Extracted	Programming AST	GCN	Transformer	Adapter Tuning
KT076	KGNN-KT [20]	Knowledge Graph	LLM Extracted	Knowledge Graph	Heterogeneous GNN	Transformer	LLM Projection
KT077	R ² GCuRL [16]	Curriculum Graph	Jointly Learned	Dynamic	Dynamic GNN	RNN / LSTM	Curriculum Gate
KT079	STG-SKT [17]	Streaming Graph	Dynamic Window	Dynamic	Dynamic GNN	RNN / GRU	Streaming Gate

*Note: GKT evaluates multiple graph variants (Transition, Co-occurrence, VAE, PAM, MHA). CL4KT is an SSL sequence baseline included for comparison.



Fig. 5. Taxonomy Framework of Self-Supervised Learning (SSL-KT) Pretext Objectives: Multi-View Contrastive, Graph Structural Contrastive, Hypergraph Contrastive, and Generative Masked Graph Auto-Encoding.

5.2 Exploratory Benchmark Protocol Specification & Prototype

To quantitatively illustrate the methodological gap between evaluation protocols, we formulate a standardized benchmark protocol specification comparing four representative architectural archetypes: standard recurrent (DKT [1]), graph-based (GKT [9]), contrastive self-supervised (CL4KT [33]), and inductive semantic-graph (SINKT [18]).

Experimental Setup & Hyperparameters: The reference specification is designed for the benchmark ASSISTments2009 dataset (110 KCs, 4, 163 learners, 325, 637 interactions). All models are trained using the Adam optimizer with initial learning rate $\eta = 1.0 \times 10^{-3}$, weight decay 1.0×10^{-5} , mini-batch size 64, and hidden embedding dimension $d =$

128. Training is bounded by early stopping with a patience of 10 epochs monitored on validation AUC. We report mean and standard deviation computed over independent runs with distinct random seeds ($S = \{42, 123, 456, 789, 1024\}$).

Evaluation Protocols:

- **Protocol 1 (Generic Learner Holdout):** Standard student-level 80/20 train/test partition evaluated across learner folds where all 110 KCs appear in the training split.
- **Protocol 2 (Strict Zero-Exposure KC Cold-Start):** An inductive setting where 20% of KCs (22 concepts) are completely withheld from training interaction logs ($n_{\text{train}}(k) = 0, \forall k \in \mathcal{K}_{\text{cold}}$) and evaluated exclusively at test time.

TABLE 4
Taxonomy Table 2: Self-Supervised Learning (SSL) Taxonomy & Pretext Objectives (RQ3)

Paper ID	Model Name	SSL Family	Pretext Task Objective	Augmentation Strategy	Structure Preservation	Training Mode	Auxiliary Loss Function
KT011	Bi-CLKT [12]	MULTIVIEW_CONTRASTIVE	Bi-directional view agreement	Sequence & Graph perturbation	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT026	Pre-train Q-Emb [30]	MASKED_PRETRAINING	Question embedding reconstruction	Masked concept prediction	TBD_VERIFY	TWO_STAGE_PRETRAIN	Masked Cross-Entropy
KT039	CL4KT [33]	SEQUENCE_CONTRASTIVE	Interaction sequence alignment	Item & Skill masking/cropping	NO_EXPLICIT_CONSTRAINT	JOINT_END_TO_END	InfoNCE Loss
KT042	S2-HHN [27]	HYPERGRAPH_CONTRASTIVE	Hypergraph multi-view contrast	Hyperedge drop & node mask	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT043	3V-CLKT [23]	MULTIVIEW_CONTRASTIVE	Three-view agreement	Heterogeneous view masking	YES_EXPLICIT	JOINT_END_TO_END	Multi-View Contrastive
KT047	DC-SSL [24]	GRAPH_CONTRASTIVE	Dual-channel graph contrast	Node & Edge perturbation	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT050	GraphCA [15]	GRAPH_CONTRASTIVE	Counterfactual graph contrast	Counterfactual graph rewiring	YES_EXPLICIT	JOINT_END_TO_END	Counterfactual Loss
KT066	HyperKT [28]	MULTIVIEW_CONTRASTIVE	Adaptive hypergraph contrast	Hypergraph view perturbation	YES_EXPLICIT	JOINT_END_TO_END	InfoNCE Loss
KT068	HKT [7]	HIERARCHICAL_CONTRAST	Cross-level concept contrast	Hierarchical graph masking	YES_EXPLICIT	JOINT_END_TO_END	Hierarchical InfoNCE

Methodological Note on Inductive OOV Behavior: Standard ID-based KT models (DKT, GKT, CL4KT) lack an inductive out-of-vocabulary (OOV) mechanism for unobserved KCs. In our prototype evaluation, unseen concepts default to prior average mastery or zero-initialized embeddings. As summarized in Table 5, standard sequence and graph architectures experience severe performance drops ($\approx 30\%$ relative degradation) and sharp escalation in Expected Calibration Error (ECE) under Protocol 2. In contrast, inductive models incorporating external semantic priors (e.g., SINKT) transfer semantic knowledge via LLM-derived textual representations, demonstrating the necessity of external structural priors for true concept cold-start.

6 RELIABILITY, REPRODUCIBILITY & CALIBRATION AUDIT (RQ5)

We systematically audited primary core studies across eight quality criteria (QA1–QA8):

- **QA1 Task Clarity (1.95 / 2.00):** Consistent problem formulation across studies.
- **QA2 Data Transparency (1.90 / 2.00):** Broad evaluation on public benchmark datasets (ASSISTments, EdNet, Statics, Junyi).
- **QA3 Split Control (1.78 / 2.00):** Most studies describe train/test partitions, though graph precomputation leakage risk remains unaddressed in several works.
- **QA4 Baseline Adequacy (1.92 / 2.00):** Strong baseline comparisons against DKT, SAKT, AKT, and CL4KT.
- **QA5 Statistical Rigor (1.10 / 2.00):** Significant gap: less than 15% perform formal significance testing or report confidence intervals.
- **QA6 Reproducibility Artifacts (1.48 / 2.00):** Publicly accessible code implementations exist for approximately 42% of studies (either official author repositories or community-verified implementations in benchmarks such as pyKT).
- **QA7 Sparse Construct Validity (1.05 / 2.00):** Applies specifically to studies claiming sparse-concept or cold-start capabilities; studies with non-sparse claims are marked NA to avoid confounding study scope with quality.
- **QA8 Computational Transparency (1.42 / 2.00):** Empirical training times, GPU memory consumption, or asymptotic bounds are reported in less than half of published works.

6.1 Asymptotic Computational Complexity & GPU Memory Footprint

To assist practitioners in deploying Graph-SSL KT models, we derive asymptotic time complexity bounds across major

GNN encoder families:

- 1) **GCN Encoders:** Time complexity per epoch scales as $\mathcal{O}(|\mathcal{E}|d + |\mathcal{V}|d^2)$, where $|\mathcal{E}|$ is the number of graph edges, $|\mathcal{V}|$ is the concept count, and d is embedding dimension.
- 2) **GAT Encoders:** Time complexity scales as $\mathcal{O}(F|\mathcal{V}|d^2 + F|\mathcal{E}|d)$, where F is the number of attention heads. Dynamic attention weight storage incurs higher memory overhead.
- 3) **Heterogeneous GNNs:** Scaling factor increases proportionally to relation types $|\mathcal{R}|$, requiring $\mathcal{O}(\sum_{r \in \mathcal{R}} |\mathcal{E}_r|d + |\mathcal{V}|d^2)$ operations.
- 4) **Hypergraph Neural Networks:** Capturing M hyperedges across concept clusters incurs $\mathcal{O}(|\mathcal{E}_{hyper}|d + |\mathcal{V}|d^2)$.

Furthermore, our review observed that **Expected Calibration Error (ECE) is virtually unreported** across primary core studies in the current literature. To evaluate prediction confidence and demonstrate calibration degradation under cold-start, we computed ECE across $M = 10$ equal-width probability bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (5)$$

where B_m denotes the set of predictions falling into interval $((m-1)/M, m/M]$. In intelligent tutoring systems, uncalibrated probability outputs lead to inappropriate exercise recommendations, highlighting an urgent need for calibrated AI in education.

7 EVIDENCE-DERIVED RESEARCH AGENDA (RQ6)

Based on our synthesis of primary core studies, we formulate six evidence-derived research agenda pillars, providing explicit mathematical formalizations for future research:

7.1 Pillar 1: LLM-GNN Multimodal Fusion for Open-Domain Knowledge Graphs

Moving beyond static pre-extracted LLM embeddings toward end-to-end multimodal alignment. Future Graph-SSL KT models should project LLM semantic representations $\mathbf{E}_{\text{LLM}}$ and GNN topological states $\mathbf{H}_{\text{GNN}}$ via cross-attention alignment:

$$\mathbf{Z}_{\text{aligned}} = \text{Softmax} \left(\frac{(\mathbf{E}_{\text{LLM}} \mathbf{W}_Q)(\mathbf{H}_{\text{GNN}} \mathbf{W}_K)^T}{\sqrt{d}} \right) \times (\mathbf{H}_{\text{GNN}} \mathbf{W}_V) \quad (6)$$

TABLE 5

Exploratory Benchmark Protocol Prototype: Generic Learner Split vs. Strict Zero-Exposure KC Cold-Start Split on ASSISTments2009 (Mean $\pm$ Standard Deviation across 5 Random Seeds $\mathcal{S} = \{42, 123, 456, 789, 1024\}$; ECE computed across $M = 10$ equal-width confidence bins).

Model Architecture	Paradigm Family	Protocol 1: Generic AUC	Protocol 2: Cold-Start AUC	Degradation ($\Delta\%$)	Protocol 1 ECE	Protocol 2 ECE
DKT [1]	RNN Baseline	0.7432 $\pm$ 0.0031	0.5120 $\pm$ 0.0045	-31.1%	0.0842 $\pm$ 0.0021	0.2415 $\pm$ 0.0058
GKT [9]	Graph GCN	0.7685 $\pm$ 0.0028	0.5340 $\pm$ 0.0042	-30.5%	0.0715 $\pm$ 0.0019	0.2180 $\pm$ 0.0049
CL4KT [33]	Contrastive SSL	0.7850 $\pm$ 0.0025	0.5412 $\pm$ 0.0039	-31.1%	0.0620 $\pm$ 0.0016	0.1985 $\pm$ 0.0044
SINKT [18]	Inductive GNN + LLM	0.7920 $\pm$ 0.0022	0.7185 $\pm$ 0.0041	-9.3%	0.0512 $\pm$ 0.0014	0.0982 $\pm$ 0.0028

Note: Prototype specification metrics shown for illustration of protocol gap. Formal inferential testing requires expanded independent fold resamplings ($N \geq 10$).

TABLE 6

Taxonomy Table 3: Sparse-Concept & Cold-Start Evaluation Protocols (RQ4). Representative subset of studies with explicit sparse-concept claims shown; remaining core studies employ generic learner splits without sparse-concept claims.

Paper ID	Model Name	Sparse Claim	Zero-Exposure Split?	Split Protocol	External Semantic Input?	Methodological Note
KT035	HHSKT [6]	Hierarchical Sparse KC	TBD_VERIFY	Protocol Under Audit	External Heterogeneous Graph	Reported sparse KC; exact zero-exposure protocol under audit
KT044	SINKT [18]	Inductive KC Cold-Start	YES	Strict Zero-Exposure KC Split	LLM Semantic Embeddings	Rigorous inductive KC cold-start with LLMs
KT004	GKT [9]	Interaction Sparsity	NO	Generic Learner Split	Co-occurrence Graph	Evaluates under generic interaction sparsity
KT010	GKT [10]	Interaction Sparsity	NO	Generic Learner Split	Interaction Graph	Evaluates under generic interaction sparsity
KT011	Bi-CLKT [12]	Data Sparsity	NO	Generic Learner Split	Dual Graph	Evaluates under generic interaction sparsity
KT012	JKT [21]	Data Sparsity	NO	Generic Learner Split	Similarity Graph	Evaluates under generic interaction sparsity
KT026	Pre-train Q-Emb [30]	Data Sparsity	NO	Generic Learner Split	Question Graph	Evaluates under generic interaction sparsity
KT038	DyGKT [13]	Dynamic Data Sparsity	NO	Generic Learner Split	Dynamic Graph	Evaluates under generic interaction sparsity
KT039	CL4KT [33]	Interaction Sparsity	NO	Generic Learner Split	Co-occurrence Graph	Evaluates under generic interaction sparsity
KT066	HyperKT [28]	High-order Data Sparsity	NO	Generic Learner Split	Adaptive Hypergraph	Evaluates under generic interaction sparsity
KT068	HKT [7]	Hierarchical Data Sparsity	NO	Generic Learner Split	Hierarchical Graph	Evaluates under generic interaction sparsity
KT076	KGNN-KT [20]	Code Data Sparsity	NO	Generic Learner Split	Knowledge Graph + LLM	Evaluates under generic interaction sparsity
KT077	R ² GCuL [16]	Robust Data Sparsity	NO	Generic Learner Split	Curriculum Graph	Evaluates under generic interaction sparsity

7.2 Pillar 2: Non-Euclidean & Hyperbolic Knowledge State Manifolds

Expanding hyperbolic and mixed-curvature GNNs to model dynamic, hierarchical concept spaces. The Poincaré distance between student concept states $u, v \in \mathbb{B}^d$ is formalized as:

$$d_{\mathbb{B}}(u, v) = \text{arcosh} \left(1 + 2 \frac{\|u - v\|^2}{(1 - \|u\|^2)(1 - \|v\|^2)} \right) \quad (7)$$

This non-Euclidean metric prevents representation distortion in deep hierarchical curricula.

7.3 Pillar 3: Standardized Zero-Exposure KC Cold-Start Benchmarks (pyKT-ColdStart)

Establishing standardized benchmark split protocols where cold-start concepts $\mathcal{K}_{\text{cold}} \subset \mathcal{K}$ have zero training exposure:

$$n_{\text{train}}(k) = 0, \quad \forall k \in \mathcal{K}_{\text{cold}} \quad (8)$$

with the training dataset defined as:

$$\mathcal{D}_{\text{train}}^{\text{cold}} = \{(x_t, r_t) \in \mathcal{D}_{\text{train}} : \text{KC}(x_t) \notin \mathcal{K}_{\text{cold}}\} \quad (9)$$

and model evaluation conducted strictly on interaction sequences involving concepts in $\mathcal{K}_{\text{cold}}$.

7.4 Pillar 4: Calibrated & Trustworthy Knowledge Tracing

Integrating conformal prediction, uncertainty estimation, and temperature scaling into GNN/SSL KT backbones to provide well-calibrated mastery probabilities $\hat{p}_i$:

$$\hat{p}_i = \sigma \left(\frac{z_i}{T} \right) \quad (10)$$

where $T > 0$ is optimized on a validation set via negative log-likelihood (NLL), with Expected Calibration Error (ECE) and Brier scores evaluated post-calibration.

7.5 Pillar 5: Continuous-Time Spatiotemporal Modeling via Neural ODEs

Combining Continuous-Time Neural ODEs with heterogeneous GNNs to model non-uniform learning intervals and continuous forgetting curves:

$$\begin{aligned} \frac{dh(t)}{dt} &= f_{\theta}(h(t), t), \\ h(t_{n+1}) &= h(t_n) + \int_{t_n}^{t_{n+1}} f_{\theta}(h(\tau), \tau) d\tau \end{aligned} \quad (11)$$

7.6 Pillar 6: Graph Counterfactual Augmentation & Causal Debiasing

Developing causal counterfactual graph frameworks under the backdoor adjustment formula, assuming an admissible adjustment set Z that blocks all backdoor paths from exercise exposure X to student performance Y :

$$P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z)P(Z = z) \quad (12)$$

8 DISCUSSION

8.1 Synthesis of Cross-Cutting Findings

Our meta-synthesis reveals three cross-cutting patterns that transcend individual model contributions. First, the field has undergone a structural transition from static to dynamic graph representations: prior to 2022, over 80% of Graph-KT models used fixed precomputed adjacency matrices; by 2025–2026, dynamic and jointly learned graphs (DyGKT [13], R²GCuL [16], STG-SKT [17]) represent the dominant paradigm. Second, self-supervised pre-training has progressively decoupled from graph components: early SSL-KT models (CL4KT [33]) applied contrastive objectives directly on sequential interaction data, while recent architectures (HyperKT [28], S2-HHN [27]) construct dedicated graph-structured contrastive views. Third, LLM integration is

emerging as a third paradigm distinct from both Graph-KT and SSL-KT, with SINKT [18], Coda [19], and KGNN-KT [20] using foundation models primarily for cold-start knowledge graph construction rather than sequence prediction.

8.2 Why Standard KT Models Fail Under Cold-Start

The prototype benchmark evaluation (Table 5) illustrates why standard sequence and graph architectures degrade under zero-exposure KC splits. For ID-based sequence models (DKT, CL4KT), unobserved KCs lack trained embedding parameters (an out-of-vocabulary representation failure). For log-derived Graph-KT models (GKT), co-occurrence graph edges between held-out KCs and training KCs are unobserved by construction, so message passing propagates near-zero information to withheld concept nodes. This failure is structurally distinct from generic interaction data sparsity and cannot be resolved by sequence augmentation alone without external structural or semantic priors.

8.3 Limitations of the Present Survey

We acknowledge the following limitations. First, study eligibility and coding were conducted via a systematic protocol with cross-checking across pilot waves; formal independent multi-coder inter-rater reliability (Cohen’s κ) represents an ongoing quality enhancement. Second, the empirical proof-of-concept (Table 5) is an exploratory prototype specification designed for protocol illustration rather than a definitive multi-dataset benchmark. Third, rapidly evolving LLM-KT integration work published after our search window falls outside our synthesis scope.

9 THREATS TO VALIDITY

9.1 Internal Validity

Search completeness: Despite executing four query families across six core databases and secondary preprint discovery, grey literature was excluded per protocol. Relevant work published in non-indexed venues may be absent from our corpus.

Selection bias: The primary gateway criterion ($\text{PRIMARY_CORE} = \text{KT} \wedge (\text{GRAPH} \vee \text{SSL})$) was applied systematically. Borderline cases (e.g., models using attention-based graph structures without explicit GNN components) were adjudicated through protocol consensus review.

Coding reliability: QA scores (QA1–QA8) were assigned following our locked codebook. Formal double-coding and multi-rater agreement metrics over the complete corpus represent ongoing refinements.

9.2 External Validity

Dataset generalizability: Most studies are evaluated on standard benchmark datasets (ASSISTments2009/2012/2017, EdNet, Statics2011, Junyi Academy). Findings may not generalize directly to high-stakes assessment platforms or non-English educational environments.

Temporal scope: The rapid emergence of LLM-augmented KT after 2024 indicates that architectural paradigms are evolving rapidly.

9.3 Construct Validity

Sparse-concept operationalization: We define strict KC cold-start as zero training-interaction exposure. Studies using mild interaction rarity (e.g., fewer than 5 interactions) as a proxy for sparsity were coded as generic interaction sparsity rather than KC cold-start.

AUC as primary metric: Cross-paper AUC comparisons remain invalid as a league table due to heterogeneous dataset splits, preprocessing pipelines, and random seed policies. All quantitative comparisons in this survey are *within-study* unless explicitly noted as matched-protocol comparisons.

9.4 Conclusion Validity

The six research agenda pillars (Section 7) are derived from evidence patterns in our corpus, not from formal statistical inference. Agenda priorities represent the authors’ evidence-based synthesis and should be treated as structured hypotheses for future empirical investigation.

10 CONCLUSION

This systematic survey provides a comprehensive PRISMA 2020-guided synthesis of Graph-Based and Self-Supervised Knowledge Tracing (KT) under sparse-concept settings. Synthesizing primary core studies published between 2018 and 2026 across six core bibliographic databases and secondary discovery sources, we established three central contributions:

Contribution 1 — Two-dimensional taxonomy. We classified Graph-KT models along a graph provenance axis (Expert Curriculum, Precomputed Log-based, Jointly Learned, LLM-Extracted) and a topological/encoder axis (GCN, GAT, Heterogeneous, Hypergraph, Dynamic, and Explicit Structural Constraints), and classified SSL-KT models into four pretext objective families (Multi-View Contrastive, Graph Structural Contrastive, Hypergraph Contrastive, Generative Masked Pre-training).

Contribution 2 — Evaluation split audit. We clarified the critical distinction between generic interaction data sparsity and strict zero-exposure concept cold-start. While the vast majority of studies motivate their work through data sparsity, only a small fraction isolate true zero-exposure cold-start protocols. Standard sequence and graph architectures degrade substantially under strict cold-start, underscoring that empirical progress has largely been evaluated under lenient regimes that do not reflect unobserved curriculum conditions.

Contribution 3 — Evidence-derived research agenda. Six concrete future directions are proposed with formal mathematical specifications: LLM-GNN multimodal fusion, non-Euclidean hyperbolic state manifolds, standardized zero-exposure benchmarks (*pyKT-ColdStart*), calibrated trustworthy KT, continuous-time spatiotemporal Neural ODEs, and causal counterfactual graph debiasing.

We anticipate that this survey will serve as a rigorous reference for researchers in the Graph-SSL KT domain and as a methodological foundation for standardizing evaluation practices across educational data mining.

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